DIF 1308

$$(e^{-x})' = (-e^{-x})$$

a) $\int -5x^4\,dx = -x^5 + C$

$$\left(-\frac{1}{x^2}\right)' = (-x^{-2})'$$

b) $\int \frac{3}{4x^3}\,dx = \frac{3}{4}\int x^{-3}\,dx = \frac{3}{4}\cdot\frac{x^{-2}}{-2} = -\frac{3}{8}x^{-2} + C$

f) $\int\left(\frac{1}{2}x^8 - \frac{3}{4}x^2 + 9\right)dx = \frac{1}{2}\int x^8\,dx - \frac{3}{4}\int x^2\,dx + \int 9\,dx = \frac{1}{2}\cdot\frac{x^9}{9} - \frac{3}{4}\cdot\frac{x^3}{3} + 9x + C$

$$= \frac{x^9}{18} - \frac{x^3}{4} + 9x + C$$

g) $\int\left(\frac{4}{\sqrt{x}} + 3e^{-2x}\right)dx = 4\int x^{-\frac{1}{2}}\,dx + 3\int e^{-2x}\,dx = \frac{4}{-x} + \frac{3e^{-2x}}{2} + C$

$$\left(\frac{e^{-2x}}{-2}\right)' = -\frac{1}{2}\left(e^{-2x}\right)' = \frac{-2e^{-2x}}{-2} = 3e^{-2x}$$

$$(8e^{-5x})' = 8(e^{-5x})' = 8\cdot -5e^{-5x}$$

$$\left(\frac{e^{-5x}}{5}\right)' = \frac{1}{5}(e^{-5x})' = \frac{1}{5}\cdot -5e^{-5x} = -e^{-5x}$$

l) $\int(2\,\mathrm{sen}(x) + 5\cos(x))\,dx = 2\int \mathrm{sen}(x)\,dx + 5\int\cos(x)\,dx = -2\cos(x) + 5\,\mathrm{sen}(x)$

m) $\int\left(\frac{2}{5x^3} - \frac{5}{2e^{5x}}\right)dx = \frac{2}{5}\int x^{-3}\,dx - \frac{5}{2}\int e^{-5x}\,dx = \frac{-1}{5x^2} - \frac{5}{2}\cdot\frac{-e^{-5x}}{5} + C$

$$= \frac{-1}{5x^2} + \frac{1}{2}e^{-5x} + C$$

$$\int e^{ax}\,dx = \frac{e^{ax}}{a}$$

b) $\int x(3x^2+1)^4\,dx$

$u = 3x^2+1$

$du = 6x\,dx$

$\frac{du}{6} = x\,dx$

$(3\cdot 4)^5 = 3^5\cdot 4^5$

$\int \frac{du}{6}(u)^4 = \frac{1}{6}\int (u)^4\,du = \frac{1}{4}\cdot\frac{u^5}{5}+C = \frac{u^5}{20} = \frac{(3x^2+1)^5}{20}+C$

d) $\int \frac{2x}{x^2+9}\,dx = \int \frac{du}{u} =$

$u = x^2+9$

$du = 2x\,dx$

$\int (5x)^{-1} =$

$= \int u^{-1}\,du = \ln|u| + C$

$= \ln|x^2+9| + C$

no directa (i)

j) $\int\left(\frac{1}{sen(x)}+\frac{2}{sec(x)}\right)dx = \int\frac{1}{sen(x)}dx + 2\int\frac{1}{sec(x)}dx$

$= \int cosec(x)\,dx + 2\int cos(x)$

$= \ln|cosec(x) - cotan(x)| + 2\,sen(x) + C$

i) $\int 2x\,sen(7x^2+3)\,dx = \int sen(u)\cdot\frac{du}{7}$

$u = 7x^2+3$

$du = 7\cdot 2x\,dx$

$\frac{du}{7} = 2x\,dx$

$= \frac{1}{7}\int sen(u)\cdot du = \frac{1}{7}\cdot -cos(u)$

$= -\frac{1}{7}\cdot cos(7x^2+3)+C$

ñ) $\int 7x^2\cdot\sqrt[3]{5x^3-2}\,dx$

$u = 5x^3-2$

$du = 5\cdot 3x^2\,dx$

$\frac{du}{15} = x^2\,dx$

$= 7\int x^2\cdot(5x^3-2)^{\frac{1}{3}}\,dx$

$= 7\int (u)^{\frac{1}{3}}\frac{du}{15} = \frac{7}{15}\int (u)^{\frac{1}{3}}du = \frac{7}{15}\frac{u^{4/3}}{4/3} = \frac{21}{60}u^{4/3}+C = \frac{7}{20}(5x^3-2)^{4/3}+C$